PIONEER JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION
HIGHER 2

CANDIDATE NAME

CT GROUP	1	0			

INDEX NUMBER				

CHEMISTRY

9647/02

Paper 2 Structured

19 September 2011

2 hours

Additional Materials: Answer Paper
Data Booklet

READ THESE INSTRUCTONS FIRST

Write your Centre number, index number and name on all work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough workings.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

FOR EXAMINER'S USE			
Paper 2			
1	/ 12	5	/ 7
2	/ 12	6	/ 4
3	/ 15	7	/ 16
4	/ 6	Total	/ 72

Answer **all** the questions in the spaces provided.

1 Planning (P)

There were two unknown solutions labelled **FA1** and **FA2,** which could either be 1.00 mol dm^{-3} sodium hydroxide or 1.00 mol dm^{-3} barium hydroxide.

A student was tasked to plan a simple experiment to identify **FA1** and **FA2** by determining the heat evolved from the reactions of each solution with 1.00 mol dm^{-3} hydrochloric acid.

She was provided with:

- a thermometer
- a polystyrene cup
- other common apparatus in the laboratory

(a) With reference to the reaction between hydrochloric acid and barium hydroxide, write an equation to represent the standard enthalpy change of neutralisation.

[1]

(b) The student proposed to carry out two experiments using the volumes given in the table and measuring the temperature rise for each experiment.

Experiment	Volume of FA1 / cm^3	Volume of FA2 / cm^3	Volume of HC*l* / cm^3	Temperature rise / °C
1	45		45	ΔT_1
2		45	45	ΔT_2

Briefly explain why she would **not** be able to identify the two solutions.

[1]

(c) **(i)** Assuming you were the student, suggest appropriate volumes of **FA1**, **FA2** and hydrochloric acid for experiment **3** and **4** which will allow you to determine the identity of the solutions.

You are required to keep the total volume of the reaction mixture at 90 cm^3.

Experiment	Volume of FA1 / cm^3	Volume of FA2 / cm^3	Volume of HC*l* / cm^3	Temperature rise / °C
3	**x**		**y**	ΔT_3
4		**x**	**y**	ΔT_4

Value of **x** = ________________ Value of **y** = ________________

(ii) Using the values of **x** and **y** you have suggested in **(c)(i)** and the following data, calculate the temperature rise for

(**I**) reaction between sodium hydroxide and hydrochloric acid

(**II**) reaction between barium hydroxide and hydrochloric acid

Hence, explain how the identity of the unknown solutions can be determined.

Standard enthalpy change of neutralisation = -57.3 kJ mol^{-1}
Specific heat capacity of the solutions = 4.2 J K^{-1} cm^{-3}

(iii) Write a plan to carry out experiments **3** and **4**. You may use the apparatus normally found in a school or college laboratory. Your plan should include essential details such as the volume of solutions and apparatus.

[7]

(d) The experiment can be carried out using a thermometer with 1°C or 0.2°C interval to measure the change in temperature for the reaction between **FA1** and hydrochloric acid.

It is known that the error (or uncertainty) that is associated with each reading when using a thermometer with 1 °C interval is ±0.5 °C, while that using a thermometer with 0.2°C interval is ±0.1 °C.

Given that the temperature rise = 6.5°C, calculate the maximum total percentage error (or uncertainty) in the temperature rise of the reaction mixture for the above experiment when using:

(i) thermometer with 1°C interval

(ii) thermometer with 0.2°C interval

[2]

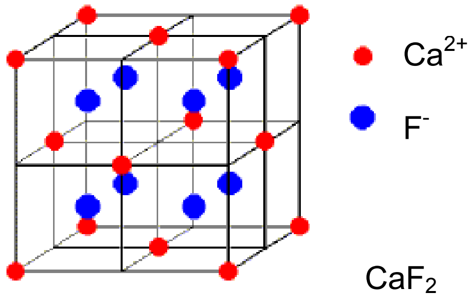
(e) Identify one potential safety hazard in this experiment and explain how you would minimise this risk.

[1]

[Total:12]

2 The major natural source of fluoride is the mineral *fluorspar*, which is mainly calcium fluoride, CaF_2.

(a) **(i)** In the ionic lattice of calcium fluoride, each calcium ion has a co-ordination number of 8 whereas fluoride ion has a co-ordination number of 4.



Explain what is meant by the term *co-ordination number* when used to refer to an ionic lattice.

(ii) Using the following information, and relevant data from the *Data Booklet*, construct an energy cycle to calculate the lattice energy of calcium fluoride, CaF_2.

Electron affinity of fluorine = -328 kJ mol^{-1}
Enthalpy change of atomisation of calcium = $+178$ kJ mol^{-1}
Enthalpy change of formation of calcium fluoride = -1220 kJ mol^{-1}

(iii) The lattice energy of a metal is small compared with that of an ionic compound. The value for the lattice energy of calcium is –178 kJ mol^{-1}. Comment on and suggest reasons for the difference between this value for calcium and the value you obtained in **(a)(ii)** for calcium fluoride.

(iv) Using the calculated value in **(a)(ii)** and the following data, calculate the enthalpy change of solution of calcium fluoride.

Enthalpy change of hydration of Ca^{2+} = −1561 kJ mol^{-1}
Enthalpy change of hydration of F^{-} = −506 kJ mol^{-1}

[8]

(b) **(i)** The solubility product for calcium fluoride in water is 3.45×10^{-11} mol^3 dm^{-9} at 25ºC. Calculate the solubility of calcium fluoride at this temperature.

(ii) Will calcium fluoride be precipitated when 10 cm^3 of 0.020 mol dm^{-3} calcium nitrate is added to 30 cm^3 of 0.010 mol dm^{-3} sodium fluoride solution at 25°C? Show how you arrive at your conclusion.

[4]

[Total: 12]

3 (a) Sodium, magnesium and sulfur are elements of the third period.

(i) Describe what you would see when these elements are burned in air or oxygen. Write equations for the reactions that occur.

Sodium

Observations:

Equation:

Magnesium

Observations:

Equation:

Sulfur

Observations:

Equation:

(ii) The oxides of sodium and sulfur resulting from the reactions in **(a)(i)**, both react with water. Write balanced equations for these two reactions and describe the colours observed when methyl orange is added to the resulting solutions.

[5]

(b) When SO_2 gas is bubbled into an orange acidified chromium-containing solution, a green solution is formed. When drops of NaOH(aq) are added to this green solution, a grey green precipitate is formed. This precipitate dissolves when an excess of NaOH(aq) is added, forming a dark green solution, containing $[Cr(OH)_6]^{3-}$.

When excess aqueous acidified hydrogen peroxide is added to the dark green solution, the original orange solution is restored.

(i) Use E^θ values from the *Data Booklet* to explain why the orange acidified chromium-containing solution turns green, when SO_2 gas is bubbled into it. Write a balanced equation for the reaction that occurs.

(ii) Give the formula of the grey-green precipitate.

(iii) Suggest the role of aqueous hydrogen peroxide.

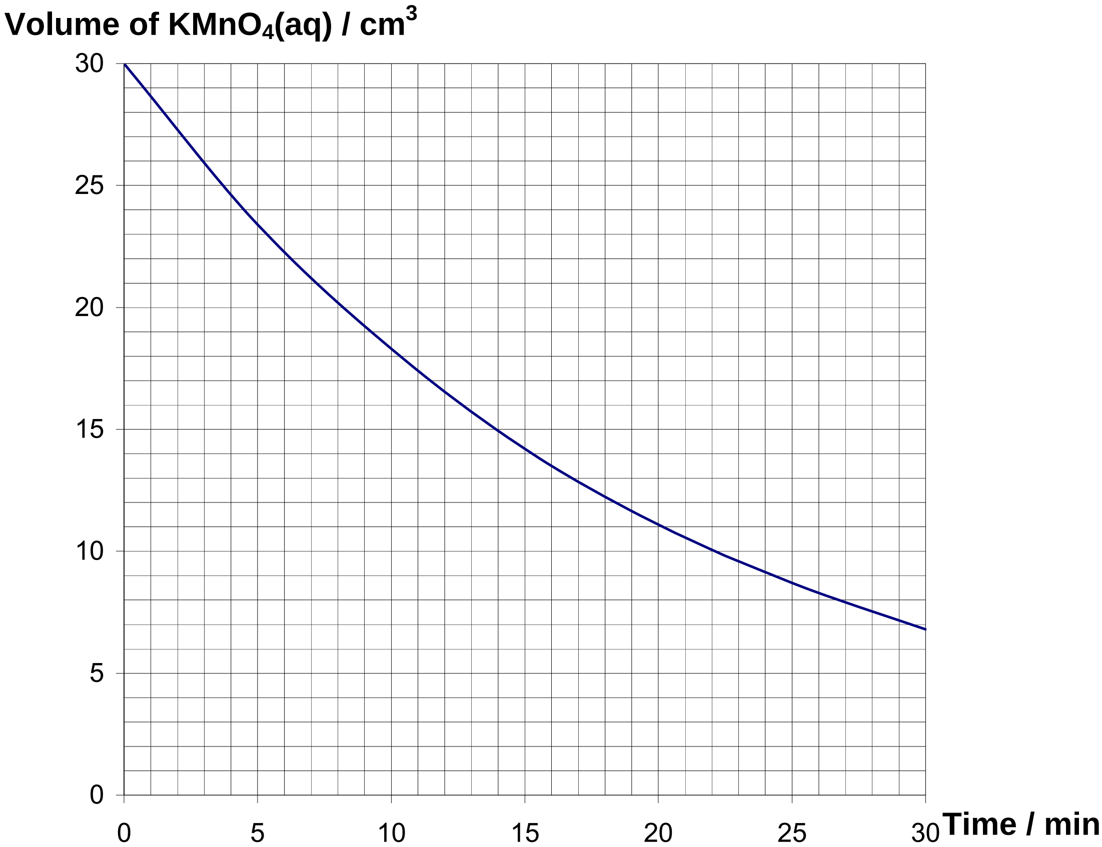
[4]

(c) A dilute solution of hydrogen peroxide decomposes slowly in aqueous solution according to the following equation:

$$2H_2O_2(aq) \rightarrow 2H_2O(l) + O_2(g)$$

A solution with an original concentration of 3.00 mol dm^{-3} was placed in a bottle contaminated with transition metal ions, which act as catalyst for the decomposition. The rate of decomposition was measured by withdrawing 10.0 cm^3 portion at various times and titrating with acidified 0.05 mol dm^{-3} $KMnO_4$(aq).

The following result was obtained.



(i) Calculate the initial concentration of hydrogen peroxide when the first portion was withdrawn.

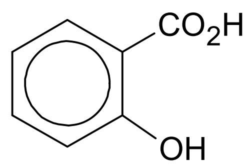
(ii) Using the graph provided, determine the order of reaction with respect to hydrogen peroxide. Show clearly how you arrive at your answer.

(iii) Hence, estimate how long the solution had been in the contaminated bottle.

[6]

[Total: 15]

4 2-hydroxybenzoic acid, **A**, is a useful intermediate for making aspirin, an analgesic.



A

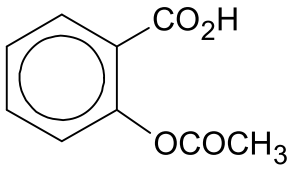
(a) The pK_a of the carboxylic acid group is 2.97. Explain how you would expect the pK_a of the phenol group to compare with that of the carboxylic acid group.

[2]

(b) The solubility of 2-hydroxybenzoic acid is 2.0 g dm^{-3}. Calculate the pH of a saturated solution of 2-hydroxybenzoic acid. You may assume that only the carboxylic acid group undergoes dissociation.

[2]

(c) Outline how aspirin may be produced from 2-hydroxybenzoic acid.

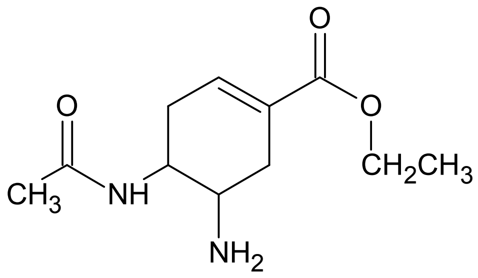


aspirin

[2]

[Total: 6]

5 Tamiflu is an antiviral drug which has been used to treat and prevent influenza A and influenza B virus infection since 1999. The structure of a derivative of Tamiflu is shown below.



Tamiflu derivative

(a) Name the functional groups present in the Tamiflu derivative.

[2]

(b) Draw the structural formula of each of the organic products formed when the Tamiflu derivative is treated with the following reagents:

(i) cold alkaline potassium manganate(VII)

(ii) hot acidified potassium dichromate(VI)

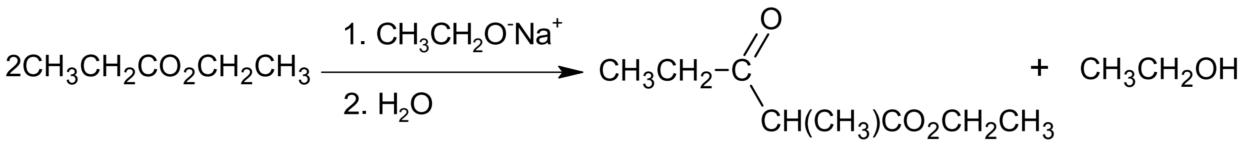
(iii) limited chloromethane with heating

[5]

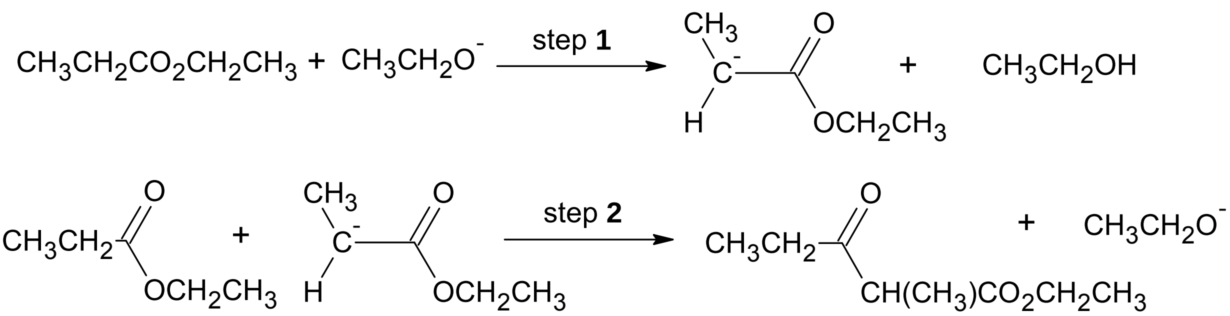
[Total: 7]

6 Claisen condensation is a carbon-carbon bond forming reaction that can occur between two esters, giving a β-keto ester. It is named after Rainer Ludwig Claisen, who first published his work on the reaction in 1881.

Claisen condensation is widely used in the biosynthesis of natural products such as fatty acids. A typical example of the Claisen condensation is the reaction of two identical esters, $CH_3CH_2CO_2CH_2CH_3$, to form ethyl-2-methyl-3-oxopentanoate.



Two simplified steps in the Claisen condensation mechanism are given below.



(a) Suggest the role of $CH_3CH_2O^-$ in step **1** of the mechanism.

[1]

(b) Suggest the type of reaction in step **2** of the mechanism.

[1]

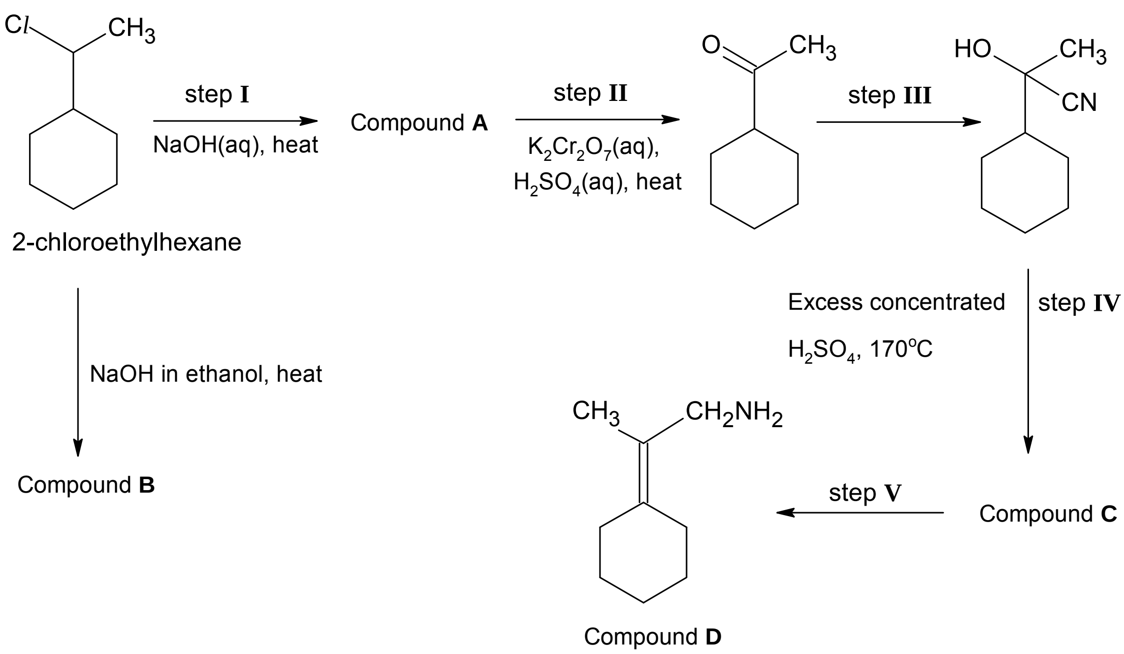
(c) Suggest the structural formulae of the final organic products formed when two molecules of $CH_3CH_2CH_2CO_2CH_2CH_2CH_3$ undergo Claisen condensation with CH_3O^- in a similar process as above.

[2]

[Total:4]

7 Amines are commonly found in polymers used in plastics or textiles. Compound **D** is an amine-containing monomer which is used in polymerisation to form plastic materials. The reaction scheme shown below outlines how compound **D** can be formed from 2-chloroethylhexane.

(a)



(i) Suggest the structural formulae for the compounds **A**, **B** and **C**.

Compound **A**	Compound **B**	Compound **C**

(ii) When optically active 2-chloroethylhexane reacted with NaOH(aq) in step **I**, the optical isomerism of the product, compound **A,** is reversed. Describe the mechanism in step **I**.

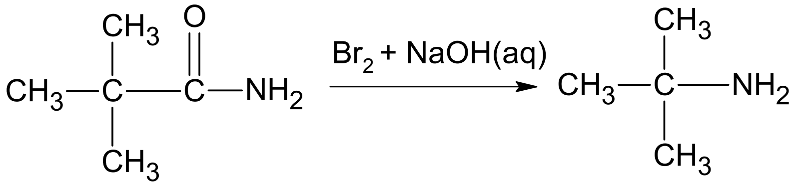
(iii) Suggest suitable reagents and conditions in step **III** and **V**.

Step **III**:

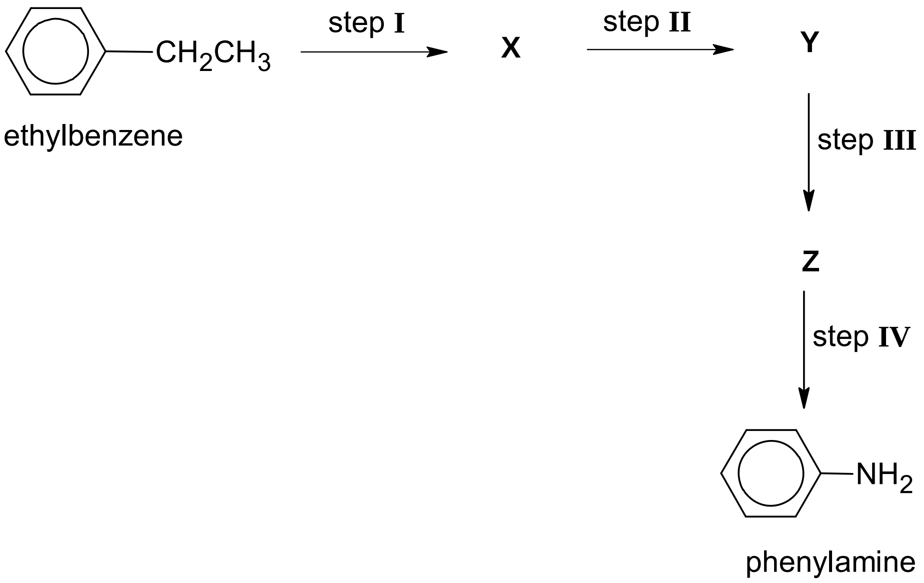
Step **V**:

[8]

(b) A method to synthesise amines directly from primary amides is the Hofmann rearrangement reaction, which was discovered by August Hofmann in 1882. Upon treatment with bromine in aqueous alkali, a primary amine is formed, as exemplified by the following:



Phenylamine can be formed from ethylbenzene in a four-step reaction sequence given below. Draw the structural formulae of the intermediate compounds **X**, **Y** and **Z** and give the reagents and conditions for step **I** to **IV**. *You are required to make use of Hofmann reaction in one of the steps.*



Compound **X**	Compound **Y**	Compound **Z**

Step **I**:

Step **II**:

Step **III**:

Step **IV**:

[5]

(c) One of the intermediates formed in Hofmann reaction has the general structure of RNCO, and is known as isocyanate.

(i) Draw a dot-and-cross diagram to show the bonding for the isocyanate intermediate, CH_3NCO.

(ii) Suggest a value for the bond angle about the nitrogen atom in CH_3NCO.

(iii) State the type of hybridisation about the two differently bonded carbon atoms.

Type of hybridisation of C in CO:

Type of hybridisation of C in CH_3:

[3]

[Total: 16]

END OF PAPER